Discover: Enzyme Action: Preparing the pH Investigation

Today I can… ☐ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. ☐ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. ☐ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Supported

1. Think of a eukaryotic animal cell. Name the part of that cell which holds the chromosomes.

2. Link to today: Think about diffusion. Which way do the particles go? Do they go from a higher concentration to a lower one, or the reverse?

3. Think about osmosis. There is a partially permeable membrane. A substance goes through it. What is the name of that substance?

4. A transport process uses energy to move particles. They go against a concentration gradient. What is the name of that process?

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Exit Ticket · Supported

Evidence

Plan: find out how pH affects amylase as it acts on starch. Iodine turns blue-black when it detects starch. It stays orange-brown when it does not detect starch. Today we plan the test for later. We have not yet collected any results from a reaction.

1. Why can amylase act on starch and then act again? Use the model or complete: 'Starch fits the ___. The enzyme is not ___.'

Word bank: active site · used up · product. Choose the words that explain binding and reuse.

2. For the pH check, name what we change, what we time and one thing we keep the same.

Use the headings Change / Time until… / Keep the same. Use the detector key to finish the timing rule.

Under the heading Exit ticket: answer the short questions in your book.